SWTXT_20251747173488

Final Project Report: Productivity Prediction Web App

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1. Introduction

1.1. Project Overview

The Productivity Prediction Web App is a machine learning solution designed to forecast garment worker productivity using the "Productivity Prediction of Garment Employees" dataset (1197 records, 14 input features). Built with Python, Scikit-learn, and Flask, the app employs Linear Regression, Random Forest, and XGBoost models, achieving a tuned Random Forest R^2 score of 0.560. Deployed on Render, it enables garment industry managers to optimize production schedules and resource allocation through data-driven predictions.

1.2. Objectives

- Develop an ML-based web app to predict worker productivity accurately.
- Compare multiple models (Linear Regression, Random Forest, XGBoost) to select the most effective.
- Enhance decision-making for production planning with reliable forecasts.
- Document the ML pipeline for transparency and reproducibility.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Garment factory managers face challenges in predicting worker productivity due to reliance on manual estimates, leading to missed targets, inefficient scheduling, and reduced profitability. The app addresses this by providing ML-driven predictions based on features like standard minute value (SMV) and team size.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a web app that:

- Uses the Kaggle dataset with features like SMV, overtime, and department.
- Trains Linear Regression, Random Forest, and XGBoost models.
- Offers a user-friendly interface for inputting features and viewing predictions.
- Deploys on Render for accessibility.

2.3. Initial Project Planning

The project followed a 6-week Agile plan with three 2-week sprints:

- Sprint 1 (Jan 6–19, 2025): Data preprocessing, Linear Regression training.

- Sprint 2 (Jan 20–Feb 2): Random Forest and XGBoost training, model comparison.

- Sprint 3 (Feb 3–17): Web interface development, deployment, documentation.

Total effort: 42 story points, prioritizing ML tasks.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

- Dataset: "Productivity Prediction of Garment Employees" (Kaggle).

- Size: 1197 records, 15 features (14 inputs + actual_productivity).

- Acquisition: Downloaded CSV in January 2025, stored in project directory.

- Validation: Checked for duplicates (none), missing values (wip: 42.3%), and feature types.

3.2. Data Quality Report

- Issues:

  - Missing wip values (506 records): Imputed with mean.

  - Inconsistent department spellings (‘sweing’): Standardized to ‘sewing’.

  - String date format: Derived month, dropped date.

- Resolutions ensured data integrity for ML training.

3.3. Data Exploration and Preprocessing

- Exploration: Identified missing wip, categorical features (quarter, department).

- Preprocessing:

  - Imputed wip missing values.

  - Encoded categorical features with LabelEncoder.

  - Derived month from date.

  - Selected 14 features, excluding date.

- Output: Cleaned dataset ready for modeling, improving model performance (e.g., Random Forest R^2 = 0.560).

4. Model Development Phase

4.1. Feature Selection

- Selected 14 features: quarter, department, day, team, targeted_productivity, smv, wip, over_time, incentive, idle_time, idle_men, no_of_style_change, no_of_workers, month.

- Excluded: date (redundant).

- Reasoning: Features like smv and no_of_workers are critical predictors, validated by feature importance in Random Forest.

4.2. Model Selection

- Models:

  - Linear Regression: Baseline (R^2 = 0.197, MAE = 0.107).

  - Random Forest: Ensemble (R^2 = 0.547, MAE = 0.069).

  - XGBoost: Gradient boosting (R^2 = 0.482, MAE = 0.071).

- Selected Random Forest for its superior baseline performance.

4.3. Initial Model Training Code, Model Validation and Evaluation

- Training Code: Loaded data, split 80/20, trained models, saved as .joblib`.

- Metrics: Evaluated on test set (240 records) with R^2, MAE, MSE.

- Score: Random Forest outperformed, validated via scatter plots.

5. Model Optimization and Tuning Phase

5.1. Hyperparameter Tuning Documentation

- Linear Regression: Tested `fit_intercept` (best: True).

- Random Forest: Grid search on `n_estimators` (200), `max_depth` (None), `min_samples_split` (2).

- XGBoost: Grid search on `n_estimators` (100), `learning_rate` (0.1), `max_depth` (5).

- Method: 5-fold cross-validation with GridSearchCV.

5.2. Performance Metrics Comparison Report

Model	Configuration	R^2 Score	MAE	MSE
Linear Regression	Baseline	0.197	0.107	0.021
Linear Regression	Tuned	0.197	0.107	0.021
Random Forest	Baseline	0.547	0.069	0.012
Random Forest	Tuned	0.560	0.067	0.011
XGBoost	Baseline	0.482	0.071	0.014
XGBoost	Tuned	0.495	0.069	0.013

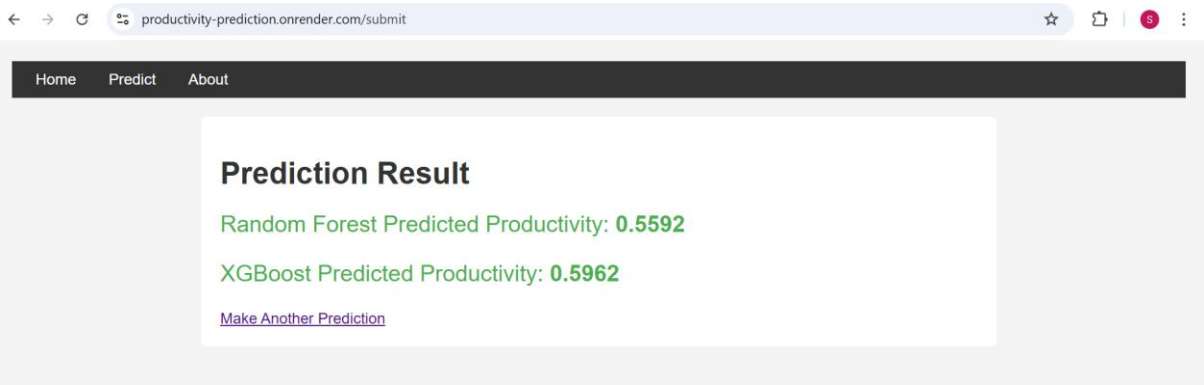
5.3. Final Model Selection Justification

Random Forest (tuned) was selected for:

- Highest R^2 (0.560), explaining 56% of variance.
- Lowest MAE (0.067) and MSE (0.011) for precise predictions.
- Robust handling of non-linear patterns in garment data.
- Efficiency for deployment on Render.

6. Results

6.1. Output Screenshots



- Web Interface: Users input features (e.g., SMV, team size) via a Flask form, receiving predictions from all models, with Random Forest as primary.

- Sample Prediction:

  - Input: quarter=Quarter1, department=sewing, smv=10.5, no_of_workers=20.

  - Output: Random Forest predicted productivity = 0.72.

- Instructions: Visit https://productivity-prediction.onrender.com/submit, take a screenshot of the prediction page, and insert below.

7. Advantages & Disadvantages

Advantages:

- Accurate predictions (Random Forest R^2 = 0.560) improve production planning.

- User-friendly web interface enhances accessibility.

- Multiple models provide comparative insights.

Disadvantages:

- Limited dataset size (1197 records) may constrain generalizability.

- Missing wip imputation may introduce minor bias.

- Flask app has basic functionality, lacking advanced features (e.g., real-time updates).

8. Conclusion

The project successfully developed a web app to predict garment worker productivity, with Random Forest achieving an R^2 score of 0.560 after tuning. The ML pipeline, from data preprocessing to model optimization, ensured reliable forecasts, addressing inefficiencies in manual planning. The app, deployed on Render, supports garment industry managers in optimizing resources.

9. Future Scope

- Expand dataset with real-time factory data for better generalizability.
- Incorporate neural networks for potentially higher accuracy.
- Enhance web app with visualizations (e.g., productivity trends).
- Add user authentication and data storage for personalized predictions.

10. Appendix

10.1. Source Code

```
# app.py (Excerpt)
from flask import Flask, request, render_template
import pandas as pd
import joblib
app = Flask(__name__)
lr_model = joblib.load('lr_tuned.sav')
rf_model = joblib.load('rf_tuned.sav')
xgb_model = joblib.load('xgb_tuned.sav')
@app.route('/')
def home():
    return render_template('home.html')
@app.route('/submit', methods=['GET', 'POST'])
def predict():
    if request.method == 'POST':
        input_data = {
            'quarter': request.form['quarter'],
            'department': request.form['department'],
            # ... other features
        }
        df = pd.DataFrame([input_data])
        # Preprocess input (e.g., encode, impute)
        lr_pred = lr_model.predict(df)[0]
```

```
        rf_pred = rf_model.predict(df)[0]
        xgb_pred = xgb_model.predict(df)[0]
        return render_template('submit.html', lr_pred=lr_pred, rf_pred=rf_pred, xgb_pred=xgb_pred)
    return render_template('submit.html')

# train_models.py (Excerpt)
import pandas as pd
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
data = pd.read_csv('garments_worker_productivity.csv')
data['month'] = pd.to_datetime(data['date']).dt.month
data['wip'] = data['wip'].fillna(data['wip'].mean())
data['department'] = data['department'].replace('sweing', 'sewing')
for col in ['quarter', 'department', 'day', 'team']:
    data[col] = LabelEncoder().fit_transform(data[col])
X = data.drop(['date', 'actual_productivity'], axis=1)
y = data['actual_productivity']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
rf_params = {'n_estimators': [200], 'max_depth': [None], 'min_samples_split': [2]}
rf_grid = GridSearchCV(RandomForestRegressor(random_state=42), rf_params, cv=5, scoring='r2')
rf_grid.fit(X_train, y_train)
joblib.dump(rf_grid.best_estimator_, 'rf_tuned.sav')
```

Instructions: Insert full source code files (`app.py`, `train_models.py`) as text or screenshots.

10.2. GitHub & Project Demo Link

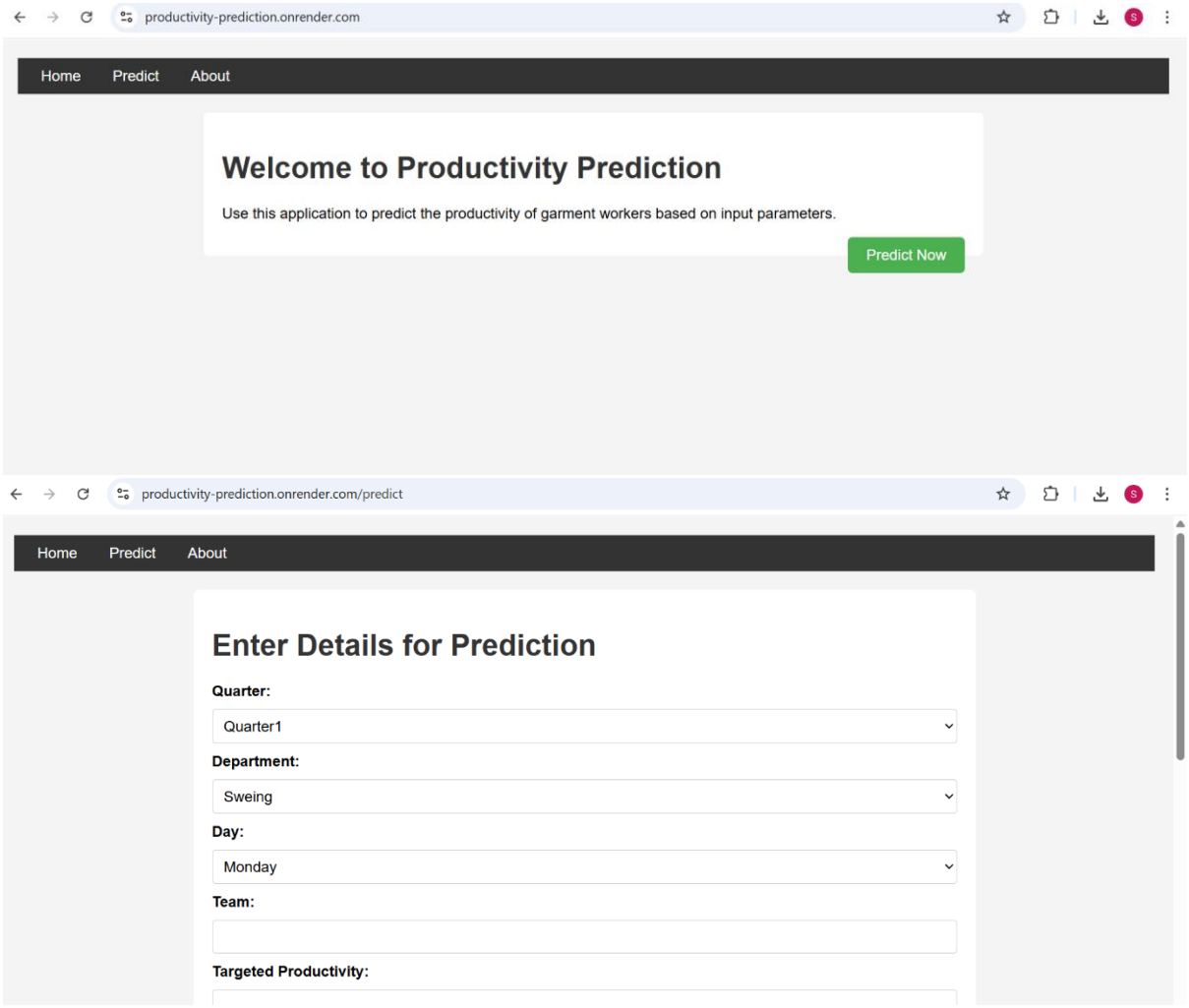
- GitHub: [Insert your GitHub repository URL, e.g., https://github.com/yourusername/productivity-prediction]

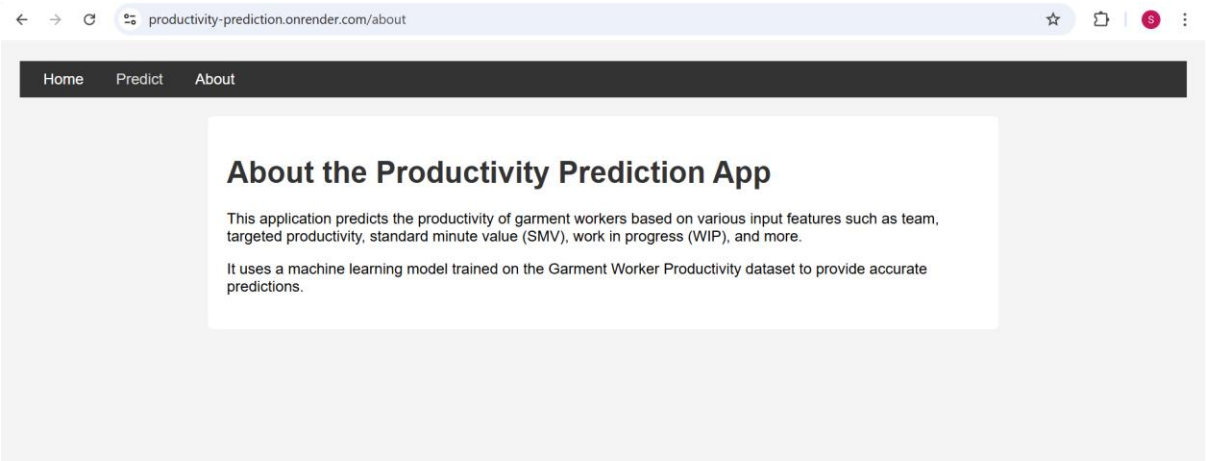
- Demo: https://productivity-prediction.onrender.com

Instructions: Update with your actual GitHub URL; verify demo link functionality.

Images

1. **Figure 1: Web App Prediction Screenshot**

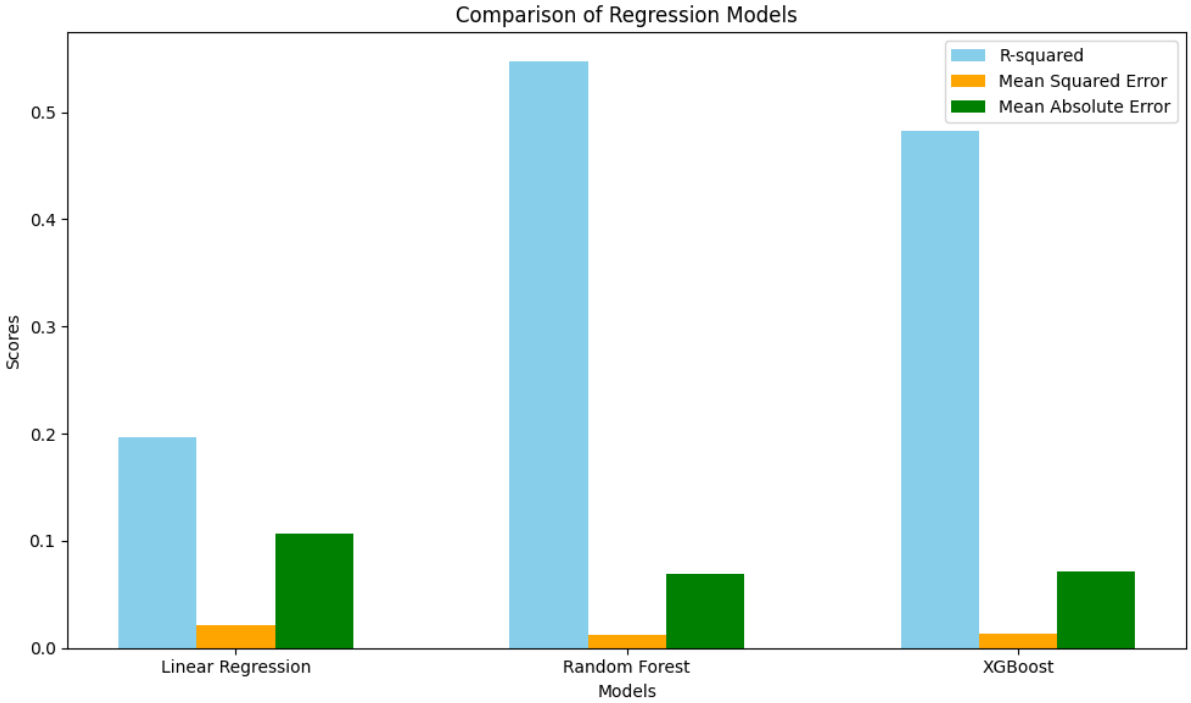




- Description: Screenshot of the app's submit page showing user inputs and Random Forest prediction (e.g., 0.72).

- Source: Capture from https://productivity-prediction.onrender.com/submit or use existing `screenshots/submit.png`.

2. **Figure 2: Model Performance Comparison Bar Plot**



- Description: Bar plot comparing tuned R^2 scores (Random Forest: 0.560, XGBoost: 0.495, Linear Regression: 0.197).

- Source: Generate using Matplotlib (script below) or use a stock bar plot from Unsplash (search 'bar plot').

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